

# JONNY LI

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## Experience

### SoundHound AI

Toronto, Canada

*Staff Machine Learning Engineer*

*Aug 2026 – Present*

- Designed and built production multi-modal LLM inference server faster than open source (vLLM, SGLang) using roofline analysis, custom Triton/CUDA kernels, CUDA graphs, continuous batching, KV cache, etc.
- Designed and implemented model evaluation and end-to-end simulation methodology for LLMs used in voice agent products, enabling data generation and post-training of custom LLMs with **90%** less cost than third party providers.

*Senior Machine Learning Engineer*

*Aug 2025 – Jul 2026*

- Built the fastest Qwen3-TTS realtime streaming inference server available today (from scratch), with input streaming and sub **40ms** first audio latency, achieving **100x** cost savings and unlocking new roadmaps for modeling teams.
- Enabled text LLMs to become voice agents through designing and implementing a low-latency realtime streaming orchestration server for ASR, LLM, TTS models, optimizing all things from low-level model inference in PyTorch/CUDA to Python asyncio event loops, reducing response latency from **2.5s to 220ms**.
- Enabled in-house post-training of LLMs up to **120B** params on long contexts larger than GPU memory by writing distributed training pipelines using Huggingface, PyTorch, DeepSpeed, Liger-Kernel, Docker, k8s, Ray, and implemented complex integrations to combine techniques such as FSDP, sequence parallelism, and PEFT.

*Machine Learning Engineer II*

*Aug 2023 – Jul 2025*

- Led ideation and experimentation of LLM-powered entity name correction for ASR by leveraging detailed data analysis, synthetic data generation, knowledge distillation, and architecture/loss function modifications to achieve **+90%** recognition accuracy on custom entity names.
- Built Bayesian optimization-based hyperparameter tuning pipeline for tuning dataset mixture weights, delivering **+30%** accuracy improvement in production ASR model.

*Software Engineer*

*Oct 2021 – Jul 2023*

- Engineered large-scale Spark ETL pipeline handling tens of TBs, debugged cluster out-of-memory issues by analyzing logical/physical plans of Spark SQL queries and implementing salting/sharding to resolve data skew across partitions, achieving **2x** throughput of previous pipeline, significantly reducing experiment iteration time.
- Authored end-to-end MLOps pipelines, designed clear and extensible APIs, resolved complex Docker dependency issues, and used Docker Compose/Kubernetes to stand up test environments for integration/end-to-end testing, which reduced manual work and experiment iteration time **5x** from **2 days to 10 hours**.

### Amazon

Vancouver, Canada

*Software Engineer Intern*

*Jun 2020 – Aug 2020*

- Built Python dependency graph analyzer to remove redundant configs, optimizing backend search engine efficiency.

## Projects (Github)

**Open Source Contributor** | *Hugging Face Transformers, TRL, PEFT*

**2024 - 2026**

- Caught and fixed several critical distributed training bug fixes to open source libraries, e.g. debugged GPU out-of-memory issues due to memory leaks and fixed several DeepSpeed ZeRO-3 integration bugs.

**Self-play RL Agents for Hexapawn** | *Python*

**2025**

- Designed custom self-play reward function and implemented training loop with classical table-based reinforcement learning algorithms like Q-learning, SARSA, and Monte Carlo on/off policy.

**CUDA Vector Search Engine** | *C++, CUDA, Python*

**2024**

- Implemented optimized vector search indices and GPU-native vector search with custom CUDA kernels, achieving **50x faster latency** vs baseline.

**Japanese Grammar Correction LLM** | *Keras, Tensorflow*

**2021**

- Implemented and trained from scratch a **state-of-the-art** grammar correction model from a Grammarly research paper and built a large-scale synthetic data generation pipeline to achieve **+10% accuracy higher than previous SOTA**.

## Education

**University of Toronto**

Toronto, Canada

*Bachelors of Science in Computer Science & Linguistics (Honours)*

*Sep 2015 – Jun 2021*

- Studied abroad at University of Tokyo from Sep 2018 - Aug 2019.